

A CERTIFIED DETERMINATION OF $\nu_3(11) = 15$ FOR ARC-DISJOINT TRANSITIVE TRIPLES IN TOURNAMENTS

ZIJIAN ZENG

ABSTRACT. For a tournament T , let $\nu_3(T)$ be the maximum number of pairwise arc-disjoint transitive triples, and let $\nu_3(n)$ be the minimum of $\nu_3(T)$ over tournaments of order n . We determine the next case of Yuster’s conjectured formula: $\nu_3(11) = 15$. For the lower bound, nauty 2.9.3 generates all 903,753,248 isomorphism classes of tournaments of order 11. Two independently written exact integer programs, using resource branching and compatibility-graph clique search respectively, find a 15-packing for every representative. A residue-by-residue verifier checks process status, exact class counts, and matching SHA-256 digests of the two regenerated input streams. The class total is also recomputed exactly from Davis’s Burnside formula. For the upper bound we use Yuster’s cyclic three-part construction with part sizes 4, 4, 3 and give an explicit 15-packing. The substantive external trust assumption is the completeness of the **gentourng** canonical enumeration; no floating-point computation, optimizer, SAT solver, or proof assistant is used.

1. INTRODUCTION

A tournament is an orientation of a complete graph. A transitive triple TT_3 is an acyclic subtournament on three vertices. A TT_3 -packing is a family of copies with pairwise disjoint arc sets. Following Kabiya and Yuster [2], write

$$\nu_3(n) = \min\{\nu_3(T) : |V(T)| = n\}.$$

Yuster conjectured

$$(1) \quad \nu_3(n) = \left\lceil \frac{n(n-1)}{6} - \frac{n}{3} \right\rceil$$

and verified it for $n \leq 8$, with the case $n = 8$ checked by computer [6, Section 4]. Kabiya and Yuster later proved the asymptotic lower bound $\nu_3(n) \geq \frac{41}{300}n^2(1 - o(1))$ using a fractional relaxation [2].

A recent public computational note determined the next two cases, $\nu_3(9) = 9$ and $\nu_3(10) = 12$, by exhaustive canonical enumeration with checkable artifacts [3]. We independently reproduce those two baselines before addressing the next case. Our result is the following.

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Theorem 1.1. *Every tournament of order 11 has a packing of 15 pairwise arc-disjoint transitive triples. Moreover, some tournament of order 11 has no packing of size 16. Consequently,*

$$\nu_3(11) = 15.$$

The proof is a certified finite computation plus a short construction. The computation concerns exactly the stated order; it neither proves the general conjecture nor classifies all order-11 minimizers.

2. DEFINITIONS AND THE HYPERGRAPH REDUCTION

Let $T = (V, A)$ be a tournament. Each unordered pair $\{x, y\}$ supports exactly one arc, so we freely index arcs by unordered vertex pairs. For a three-set $X \subseteq V$, let

$$e(X) = \binom{X}{2}.$$

The set X is transitive precisely when some vertex of X beats the other two, or equivalently when its three internal outdegrees are 0, 1, 2.

Lemma 2.1 (Exact reduction). *Let H_T be the 3-uniform hypergraph whose vertex set is $\binom{V}{2}$ and whose hyperedges are the sets $e(X)$ for transitive three-sets X . Then $\nu_3(T)$ is the matching number of H_T .*

Proof. Two copies of TT_3 share an arc exactly when their sets of underlying unordered pairs intersect. Thus a family of pairwise arc-disjoint copies is exactly a matching of H_T . $\square$

Relabeling V induces an isomorphism of the associated hypergraphs. Hence $\nu_3(T)$ is an isomorphism invariant, and one representative of every tournament isomorphism class suffices.

Two further reductions describe a hypothetical obstruction. If $\mathcal{P}$ is a maximal packing and $L(\mathcal{P})$ is the graph of unused pairs, every triangle of $L(\mathcal{P})$ must be cyclic in T ; otherwise that triple could be added. In particular $L(\mathcal{P})$ is K_4 -free, since every tournament on four vertices contains a transitive triple. At order 11, a packing of size q has $55 - 3q$ leave edges.

There is also an exact deletion criterion. Fix v and a 12-packing in $T - v$. Mark an unused pair ab when $T[\{v, a, b\}]$ is transitive. The packing extends by three triples through v if and only if the marked leave graph has a matching of size three: the matching condition is precisely what makes the six arcs incident with v distinct. Since a nine-edge graph on ten vertices need not have a 3-matching, the certified order-10 result alone does not force the order-11 result.

3. THE UPPER BOUND AND AN EXPLICIT MINIMIZER

Partition eleven vertices into V_1, V_2, V_3 of sizes 4, 4, 3. Orient every cross-part arc cyclically,

$$V_1 \longrightarrow V_2 \longrightarrow V_3 \longrightarrow V_1,$$

and orient arcs inside the parts arbitrarily. A three-set meeting all parts induces a directed cycle, so every transitive triple contains a within-part arc. A packing cannot reuse such an arc, and there are

$$\binom{4}{2} + \binom{4}{2} + \binom{3}{2} = 15$$

within-part arcs. This is Yuster’s general upper-bound construction [6, Section 4], specialized to $n = 11$.

For a literal certificate, use vertices $0, \dots, 10$, parts $0123 \mid 4567 \mid 8910$, and the following upper-triangle bit string; at position (i, j) , $i < j$, a 1 denotes $i \rightarrow j$:

11111110001111110001111100011110001111111111111111111111111111111111

The following fifteen vertex triples form a packing:

$$\begin{aligned} &(0, 1, 8), (0, 2, 9), (0, 3, 10), (1, 3, 9), (1, 2, 10), (2, 3, 8), \\ &(0, 4, 5), (0, 6, 7), (1, 4, 6), (1, 5, 7), (2, 4, 7), (2, 5, 6), \\ &(4, 8, 9), (5, 8, 10), (6, 9, 10). \end{aligned}$$

The standalone verifier reconstructs the tournament, checks these 45 distinct arcs, and checks every one of its 117 transitive triples against the within-part hitting set. This proves that the displayed tournament has packing number exactly 15 and hence that $\nu_3(11) \leq 15$.

4. THE EXHAUSTIVE LOWER BOUND

4.1. Canonical enumeration. We use `gentourng` from `nauty 2.9.3`. The `nauty` framework and its canonical-labeling methods are described by McKay and Piperno [4]. The generator writes a tournament as the $\binom{n}{2}$ upper-triangle bits in row-major order and supports disjoint residue slices. The complete order-11 run uses all residues modulo 192.

The known number of unlabeled tournaments is OEIS A000568 [5]. Independently of that table and of `nauty`, a standard-library Python program evaluates Davis’s Burnside formula [1] with exact rational arithmetic and obtains

$$a(11) = 903,753,248.$$

The 192 generated slice counts sum to this integer exactly.

4.2. Builder: branching on pair resources. For each input tournament, the Builder enumerates all transitive triples as three-bit masks in a 55-bit word. A state consists of a partial matching, its used pairs, and pairs declared permanently uncovered. It chooses a live pair e lying in the fewest available triples. Every extension either uses one available triple containing e or leaves e uncovered; the program branches over precisely these cases. A pair lying in no available triple is closed automatically. If d triples have been chosen and only L pairs remain live, the branch is rejected only when

$$d + \lfloor L/3 \rfloor < 15.$$

Lemma 4.1 (Builder soundness and completeness). *On a well-formed tournament input, the Builder returns success only with a valid 15-packing. If it exhausts the search without success, no 15-packing exists.*

Proof. Every accepted triple was enumerated as transitive, and the used-pair mask forbids intersections. The returned witness is checked again before the input is accepted. For completeness, fix any extension of a state and the chosen live pair e . Since an extension is a matching, it either contains exactly one available hyperedge through e or contains none. These are the program’s cover branches and its uncovered branch. Closure removes only a pair occurring in no available hyperedge. Finally, each additional hyperedge consumes three live pairs, proving the pruning inequality. An induction on the remaining live pairs establishes exhaustiveness. $\square$

4.3. Certifier: a different compatibility-graph search. The Certifier was written independently and imports no Builder output. It defines transitivity by the internal outdegree multiset $\{0, 1, 2\}$. Its vertices are the transitive three-sets; two vertices are adjacent when their three pair masks are disjoint. By Lemma 2.1, a target packing is exactly a clique of size 15. The Certifier orders these vertices and uses exhaustive include/exclude recursion on a three-word candidate bitset, intersecting with the later-neighbor set after inclusion. It prunes only when fewer than the required number of candidates remain.

Lemma 4.2 (Certifier soundness and completeness). *The Certifier accepts an input tournament if and only if its compatibility graph contains a clique of size 15, equivalently if and only if the tournament has a 15-packing.*

Proof. Adjacency is pair-disjointness, so the clique equivalence is immediate. At each recursion node, the first candidate is either included or excluded. In the include branch, intersection with its later neighbors retains exactly the candidates that can extend the clique without producing duplicate orders. These two cases partition all possible cliques. Candidate cardinality is an exact upper bound on how many further vertices can be included. $\square$

4.4. Complete result.

Proposition 4.3 (Certified finite sweep). *Both exact programs accept every one of the 903,753,248 canonical order-11 representatives produced by the complete residue partition. For each of the 192 residues, the programs process the same number of tournaments and report the same SHA-256 digest of the regenerated input stream.*

The strict aggregator requires one successful summary for every residue, checks the endpoint and executable identity, adds all counts, and rejects any missing, malformed, or nonzero-status slice. The resulting Builder and Certifier certificate indices are then compared residue by residue. The two scanner executables have different SHA-256 digests.

The lower bound in Theorem 1.1 follows from Proposition 4.3, isomorphism invariance, and the canonical enumeration contract of **gentourn**. The upper bound was proved in Section 3.

5. BASELINE REPRODUCTION AND ADVERSARIAL CHECKS

Before the new endpoint was attempted, both programs reproduced the known finite frontier. The target column is the right-hand side of (1).

n	isomorphism classes	target
3	2	0
4	4	1
5	12	2
6	56	3
7	456	5
8	6,880	7
9	191,536	9
10	9,733,056	12
11	903,753,248	15

The $n = 9, 10$ runs independently reproduce the exact 2026 baseline of Kirtchakov [3]; the construction of Section 3 supplies the matching upper bound at each listed order.

The Breaker route separately tested all 32 cyclic $4 + 4 + 3$ blow-ups formed from canonical within-part tournaments and 100,000 deterministic random labeled tournaments (seed 20260830). These diagnostics found no tournament below target and are not used in the universal proof. A full exact maximum search on the displayed minimizer returned 15.

Twenty-one deliberate corruptions exercised the fail-closed behavior. The scanners rejected short and nonbinary lines; the sweep aggregator rejected a missing slice, nonzero exit, malformed summary, wrong exact total, and role mismatch; the sweep comparison rejected a non-PASS certificate, altered stream digest, altered slice count, identical scanner digest, wrong endpoint, and malformed JSON; and the minimizer verifier rejected invalid JSON, a missing field, a flipped cross-part arc, a duplicate triple, a cyclic triple, and an out-of-range vertex. All negative controls were rejected.

6. TRUSTED BASE, LIMITATIONS, AND REPRODUCIBILITY

The substantive external trust assumption is that **gentourn** enumerates exactly one representative of every tournament isomorphism class. We support, but do not formally verify, that implementation claim by using all residue classes, reproducing all smaller class totals, matching the exact Burnside count at $n = 11$, and regenerating every input stream independently for the Certifier. Count agreement alone would not rule out a duplicate class paired with a missing class; canonical-generation correctness therefore remains explicitly in the trusted base.

The remaining trusted components are the C compiler, Python interpreter, macOS process and integer semantics, and SHA-256 for integrity binding. The searches themselves use only integer and bitwise operations. No proof assistant was used. We prove no classification of all equality tournaments and make no claim for $n \geq 12$.

The decisive environment was macOS 26.5 on arm64, nauty 2.9.3, Apple clang 21.0.0, and CPython 3.14.7. From the project root, the principal clean replay commands are

```
make -C code
python3 code/burnside_count.py
python3 tests/test_small_baseline.py
code/run_sweep.sh builder 11 15 192 12 \
    903753248 n11_builder_full_m192
code/run_sweep.sh certifier 11 15 192 24 \
    903753248 n11_certifier_full_m192
python3 code/compare_sweeps.py \
    certificates/n11_builder_full_m192.json \
    certificates/n11_certifier_full_m192.json \
    certificates/n11_sweeps_match.json
python3 code/verify_minimizer.py \
    certificates/minimizer_11.json
python3 tests/test_fail_closed.py
python3 tests/test_sweep_fail_closed.py
python3 tests/test_compare_fail_closed.py
```

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

The release manifest binds the formal statement, C and Python sources, per-slice summaries, aggregate certificates, explicit minimizer, audit records, LaTeX source, and PDF. The individual per-class witnesses are not stored; independent verification regenerates every canonical input and reruns the different exact decision procedure.

The research, code generation, computation orchestration, verification, auditing, and manuscript drafting were carried out in a single OpenAI Codex CLI research session under the author’s supplied execution contract. No subagents were used. Builder, Breaker, Certifier, and Referee roles were performed serially and recorded separately. The named author is solely responsible for the mathematical claims and release decision.

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INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNIVERSITY,
KUALA LUMPUR, 56000, MALAYSIA

Email address: `zijianzeng@foxmail.com`

Email address: `1002266693@ucsiuniversity.edu.my`